European Journal of Operational Research xxx (xxxx) xxx

Contents lists available at ScienceDirect

European Journal of Operational Research

journal homepage: www.elsevier.com/locate/eor

Innovative Applications of O.R.

Driving from afar: The vehicle routing problem with remote drivers

Pedro Munari [a], Nils Boysen [b,*]

[a] Production Engineering Department, Federal University of Sao Carlos, Brazil
[b] Lehrstuhl für Operations Management, Friedrich-Schiller-Universität Jena, Germany

ARTICLE INFO

Keywords:
Transportation
Vehicle routing
Remote driving
Benders decomposition

ABSTRACT

Autonomous driving is about to change the transportation of people and goods. However, due to very high safety requirements, even in exceptional situations, autonomous driving may take longer than anticipated to reach the necessary technological maturity and finally be approved for use on public roads. Remote driving, in which humans in the loop teleoperate vehicles, can be a key bridging technology. Even when autonomous driving is the standard, remote human supervisors will remain crucial for solving problems in exceptional situations. In this context, this paper extends the classical vehicle routing problem with time windows (VRPTW) by the specificities of remote driving. Remote drivers need not remain in their vehicles during service times at customer locations. Instead, they can switch to other vehicles, opening the possibility of operating a fleet with fewer drivers. We model a basic version of the resulting VRPTW with remote drivers (VRPTWRD) as a mixed integer program and resort to Benders decomposition to derive effective branch-and-Benders-cut approaches. Our computational experiments analyze both algorithmic performance and managerial insights, including changes in routing costs, driver utilization measured through idleness rates, and operational flexibility quantified by driver switches. The results reveal that remote driving can be an effective lever to reduce the workforce in times of human driver shortages.

1. Introduction

Autonomous driving has the potential to transform the future mobility of people and goods. However, while tremendous progress has been made in recent years (for a detailed overview of the development status of autonomous driving see, e.g., Teng et al., 2023), the initially high expectations for a rapid realization of autonomous driving have been delayed (e.g., Malik et al., 2023). As a result, remotely operated vehicles can play a critical role, especially during the transition period until fully autonomous driving becomes a reality. Even when fail-safe autonomous driving is eventually achieved, remote supervision, where a human operator monitors multiple autonomous vehicles in parallel and intervenes in unforeseen situations, will remain important.

Remote vehicle operation is a type of teleoperation that allows a vehicle to be controlled by a human operator without direct physical access, typically through the use of wireless communication technologies (Amador et al., 2022). Although the concept is not new, recent advances in 5G wireless cellular networks, sensor technology, high-quality cameras, and machine learning have made remote vehicle operation more feasible and successful for a wider range of applications, both on public roads and inside facilities such as warehouses and factories (Majstorović et al., 2022). There are different modes for remote driving: (i) Remote operators can drive vehicles throughout the entire operation. In this case, vehicles are non-autonomous and require active remote driving during the whole driving time. Alternatively, remote drivers can be employed to support autonomous vehicles either (ii) by intervening in specific (predefined) situations (e.g., lifting hazardous materials into a high-bay rack with a teleoperated forklift, which otherwise drives autonomously), or (iii) by resolving unforeseen circumstances (e.g., when passing a confusing construction site in bad weather). This paper addresses case (i).

Remote operation (i.e., in all three modes (i) to (iii)) has been successfully used for many years in a variety of applications, especially for environments inaccessible or too dangerous for on-board drivers. Examples are search and rescue missions, military reconnaissance, and mining (Majstorović et al., 2022). Other successful applications are reported for container ports and their teleoperated quay cranes (Kim, 2006). In recent years, several companies have entered the market of remotely operated vehicles for a wider audience. Two examples of remote-controlled forklifts and taxis from Phantom Auto and Vay, respectively, are shown in Fig. 1. *Vay* has operated remote-controlled electric cars on public roads in Germany for over three years but with a safety driver on board. As of December 2022, they conduct tests in

* Corresponding author.
E-mail addresses: munari@dep.ufscar.br (P. Munari), nils.boysen@uni-jena.de (N. Boysen).
URLs: https://www.dep.ufscar.br/munari/ (P. Munari), https://www.om.uni-jena.de (N. Boysen).

https://doi.org/10.1016/j.ejor.2026.04.050
Received 22 March 2025; Accepted 28 April 2026
Available online 2 May 2026
0377-2217/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

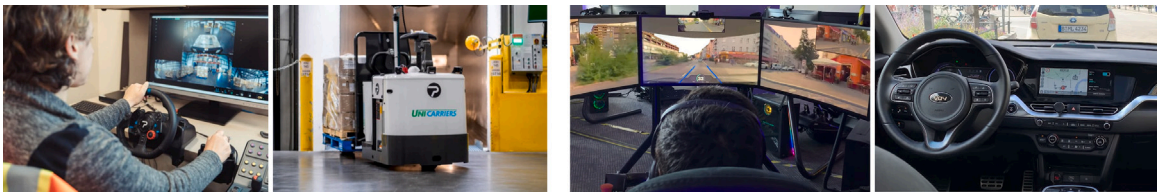


Fig. 1. Applications of remote driving. Left two pictures: Remote driver controlling a remotely operated forklift (Source: Phantom Auto https://www.phantom.auto) Right two pictures: Remote driver controlling a remote taxi (Source: Vay https://youtu.be/yifXupMSWRQ).

predefined areas without a safety driver in the car (see https://vay.io/). Other examples are Ottopia, offering software for human supervisors in the loop to support autonomous cars in unclear situations (see https://ottopia.tech/products and Amador et al., 2022), and Plus One Robotics, promoting software to remotely support autonomous picking robots in warehouses (see https://www.plusonerobotics.com/products/yonder).

Compared to traditional vehicles with on-board drivers, remote systems act as a lever to reduce labor costs (for a more detailed discussion of the pros and cons, see Amador et al., 2022; Guo et al., 2019; Yu & Lee, 2022). Remote drivers have the flexibility to switch between far-flung tasks without traveling in between, and they only need to be with a vehicle when required. Taxi drivers in New York, for example, spend about 40% of their time without any passenger on board (Zhang, 2020), and many truck drivers must wait several hours before being processed at crowded distribution centers (Boysen & Fliedner, 2010). During these vehicle waiting times, remote drivers need not wait but can switch to other driving duties. In addition, remote driving eliminates the need to commute to work and allows work to be done from regions with lower labor costs. Thus, remote driving can be an effective lever to reduce labor costs in times of human driver shortages, until autonomous driving is finally realized. Even when autonomous vehicles can safely handle any situation without human intervention, specific vehicles may still require remote human interaction, e.g., with law enforcement officers or health personnel (Zhang, 2020).

Existing research on remote-controlled vehicles mainly focuses on the technical aspects (see Amador et al., 2022; Majstorović et al., 2022), while our literature review in Section 2 will show that operations research literature on this topic is scarce. To fill this gap, this paper extends one of the most basic and elementary routing problems, the vehicle routing problem (VRP) with time windows (VRPTW), by the specifics of remote drivers. We call the resulting optimization problem VRPTWRD. Here, each drive of a vehicle along an arc needs the attendance of a unique remote driver during the whole time. Once a customer is reached and the vehicle is serviced (e.g., loaded or unloaded), the remote driver can switch to other driving duties without travel, requiring only a brief setup time for establishing a reliable remote connection. Existing vehicle routing models that explicitly consider on-board drivers (e.g., Domínguez-Martín et al., 2018; Moreno et al., 2020; Wen et al., 2021) can be applied to model remote drivers as well. However, as discussed above, this neglects their additional flexibility (i.e., switching between remote jobs without travel time) and leads to worse planning results. Therefore, adaptations of existing operations research approaches are needed to account for the additional flexibility of remote drivers.

In the VRP context, service times at customer locations (e.g., waiting time while trucks are loaded or unloaded) are the main lever for deriving more efficient schedules. While on-board drivers remain at their vehicles during service times, remote drivers can switch to other driving duties in the meantime, and another driver can take over the vehicle at the end of the service time. This add-on flexibility opens up the possibility of operating a fleet with a smaller workforce of remote operators. In this context, the main contributions of this paper are as follows:

- This paper is the first to extend a classical vehicle routing problem by the specifics of remote driving. Specifically, we formulate (a basic version of) the VRPTWRD, resolving the tradeoff between efficient vehicle tours and the number of remote drivers as follows: we fix the vehicle fleet as well as the number of available remote drivers and minimize the routing costs. Naturally, this is just one option to capture this tradeoff, but we leave versions of the VRPTWRD, including the minimization of total costs or the size of the remote driver workforce, to future research. We contribute with a mixed integer programming (MIP) model for our version of the VRPTWRD.
- To solve VRPTWRD, we apply Benders decomposition (Moreno et al., 2019; Rahmaniani et al., 2017) to the proposed MIP model and develop effective branch-and-Benders-cut (BBC) procedures that rely on existing and novel valid inequalities, different solution strategies for the subproblems, and several algorithmic enhancements.
- In our computational study, we evaluate the performance of our solution approaches and provide managerial insights, exploring the tradeoff between efficient vehicle tours and the number of remote drivers. We identify situations where remote driving can reduce idleness and save significant wage costs compared to on-board driving, and the corresponding impact on operational complexity regarding the number of vehicle switches per driver.

We organize the rest of the paper as follows. Section 2 reviews the relevant literature. In Section 3, we introduce our VRPTWRD along with a novel MIP model. In Section 4, we detail our BBC approach for the VRPTWRD, including different solvers for the subproblem of driver scheduling for given vehicle routes. Section 5 reports the results of our computational study, and Section 6 concludes the paper.

2. Literature review

Our literature review focuses on two areas. First, we survey the literature on remote driving. Then, since there is no literature on vehicle routing with remote drivers, we review related routing problems with similar problem structures.

(a) Remote Driving: The Society of Automotive Engineers (SAE) (2018) has defined six levels of vehicle autonomy, from fully manual at level 0 to fully automated at level 5. Various surveys of the status quo of autonomous driving (e.g., Khan et al., 2022; Malik et al., 2023) state that reaching level 5, a truly autonomous vehicle that can handle any situation without human intervention, is still years away. Until then, remote driving is seen as an important bridging technology that eliminates the need for a passenger to take over in situations that pre-level 5 autonomous vehicles cannot handle. Furthermore, behavioral research has shown that remote drivers in the loop promote acceptance of autonomous driving among passengers (Li et al., 2024) and drivers (Li et al., 2023). While there is a large body of operations research literature on all aspects of autonomous driving, such as route definition, maneuver planning, and providing collision-free trajectories (see the survey of Garrido & Resende, 2022), this finding does not hold for remote driving. Most research on remote driving is related to its technical implementation. Examples include

user interface design for remote drivers (Chen et al., 2007), reliability of the wireless communication link (Opiyo et al., 2021), and latency issues in wireless control (Zhang, 2020). The only operations research papers we know of are Gohar and Lee (2020) and Zulqarnain and Lee (2021). These papers address latency issues in remote driving, i.e., the time delay in transmitting signals between the vehicle and the remote driver, and the safety implications if the latency becomes too large. To do so, Gohar and Lee (2020) consider a single vehicle and multiple remote drivers with different locations, whose latency varies along the vehicle's given tour. They apply simple priority rules to assign drivers to route segments to minimize the number of drivers without violating latency constraints. In Zulqarnain and Lee (2021), this research is extended when the (single) vehicle can choose among multiple paths to its destination. We extend this research by considering multiple vehicles as well as time windows and by applying sophisticated optimization instead of simple priority rules. We follow the majority of the research reviewed by Opiyo et al. (2021) and Zhang (2020) and assume that modern wireless network infrastructure, such as 5G, makes latency differences among remote drivers a non-issue.

(b) Routing problems with similar problem structure: We assume that each drive of a vehicle along an arc is controlled by a remote human driver, who cannot switch underway while driving for safety reasons. Therefore, we need to coordinate the vehicles with the remote drivers and thus the VRPTWRD falls into the category of VRPs with synchronization (Drexl, 2012; Soares et al., 2024). Specifically, the VRPTWRD requires what this area calls *movement synchronization*. This type of synchronization "concerns the inter-route dependency between two vehicles whose task sequences need to be synchronized" (Soares et al., 2024) and is especially needed when a vehicle that cannot move autonomously needs another vehicle (in the VRPTWRD, a remote driver) to transport it. The most famous problem classes in this area are the Active-Passive VRP (Meisel & Kopfer, 2014; Tilk et al., 2018) and the Truck and Trailer Routing Problem (TTRP) (Chao, 2002; Rothenbächer et al., 2018). The former focuses on pickup and delivery tasks, thus the VRPTWRD is particularly related to the latter. The VRPTWRD can be seen as a special case of the TTRP, where the truck (the remote driver in the VRPTWRD) has zero load capacity and travels at infinite speed without the trailer (the vehicle in the VRPTWRD). Thus, without the fixed setup time for establishing a reliable remote connection that we additionally consider, the formulations and solution methods for the TTRP (for a recent survey, see Soares et al., 2024) would be directly applicable to the VRPTWRD. However, even if applicable, these approaches would suffer from significant overhead (i.e., for unnecessarily tracking the tours of the remote drivers). The flexibility of the remote drivers to switch driving tasks without traveling, requiring only a short setup time, is also the main difference from other combined vehicle and crew routing problems (e.g., Domínguez-Martín et al., 2018; Lam et al., 2020; Wen et al., 2021), where crews can also travel as passengers on-board of vehicles driven by other crews.

We conclude from our literature review that the VRPTWRD has not been treated before and requires new tailored solution methods because related routing problems are burdened by significant overhead and still cannot directly cover our problem without adaptation.

3. Problem definition of VRPTWRD

In the VRPTWRD, we are concerned with two main decisions: vehicle routing, which refers to the definition of feasible routes of the vehicles to satisfy each customer demand, and driver scheduling, which refers to the assignment of drivers to travel legs in those routes. These two decisions face a fundamental tradeoff: to save on labor costs, we can reduce the number of remote drivers. However, doing so increases the likelihood that vehicles will have to wait for remote drivers, thereby deteriorating the efficiency of vehicle routes (and vice versa). Of course, there are different ways to solve this tradeoff (e.g., by minimizing the total costs of routing, vehicles, and labor). To introduce the VRPTWRD to the operations research community, we opt for a fundamental problem version: we assume a given vehicle fleet operated by a given workforce of remote drivers and aim to minimize the routing cost. With a given truck fleet, owned by the planning company and not rentable to third parties on short notice, solutions using fewer vehicles can barely reduce costs. Analogously, leaving parts of a fixedly employed remote driver workforce, unusable for alternative duties, e.g., due to contractual or legal issues, without driving duties, barely has a cost impact. Hence, this leaves variable driving costs associated with the driven truck legs (arcs). Our VRPTWRD aims to minimize the sum of these arc-related costs, and we leave alternative objectives (e.g., minimizing total costs or the remote driver workforce) to future research.

3.1. Problem description

The VRPTWRD is defined over a network $\mathcal{G} = (\mathcal{N}, \mathcal{A})$, where $\mathcal{N} = \{0, 1, \dots, n, N\}$ is the set of nodes and $A = \{(i,j) \in \mathcal{N} \times \mathcal{N} : i \neq j, i \neq N \text{ and } j \neq 0\}$ is the set of arcs connecting these nodes. Node set $\mathcal{N}$ includes the set of customer nodes $C = \{1, \dots, n\}$ and the depot, represented by two (virtually separated) depot nodes 0 and $N = n + 1$. We consider a homogeneous fleet of K vehicles available at the depot, which can be used to visit customer nodes and satisfy their demands. Each vehicle has a one-dimensional load capacity Q and can execute at most one route, departing from the depot node 0 and returning to the depot node N after visiting a subset of customer nodes. Note that we have no explicit restriction on maximum tour lengths. However, implicitly, the capacity constraints and the time windows prevent extraordinarily long tours for single vehicles. Each customer has to be visited by exactly one vehicle. We associate a travel cost $c_{ij} > 0$ and travel time $t_{ij} > 0$ to each arc in $\mathcal{A}$ and assume that the corresponding matrices satisfy the triangle inequality.

The visit of a node $i \in \mathcal{N}$ is associated with a service time s_i, during which the vehicle must remain in the node, e.g., for being loaded and/or unloaded. Here emerges the first difference between the VRPTWRD and other VRP variants: remote drivers are not fixedly assigned to their trucks and need not wait for the vehicle during service times. Instead, they can be reassigned to another vehicle, and a different driver may (but need not) resume the route from node i. We assume that a transfer time f_i applies to node i, corresponding to the setup required by the system to establish a safe and reliable connection between the remote driver and the vehicle. Only after f_i has passed, the driver can start driving the next vehicle. Thus, we assume that — due to safety issues — remote drivers can switch driving duties only at customer nodes but not underway while driving. All available remote drivers are assumed to be identical and do not vary according to latency restrictions (see Section 2). Furthermore, we do not consider break and rest regulations (see, e.g., Goel, 2018) and assume that the drivers are available during the complete planning horizon, which is typically a working day.

We assume that H remote drivers are available and thus define the set of available drivers $\mathcal{H} = \{1, \dots, H\}$. Notably, while vehicles are assigned to at most one route, drivers may be assigned to several different routes throughout their duty, but only one at a time. Hence, at each point in time, at most H vehicles can be in transit, but we may have more than H routes in execution.

As in traditional variants of the VRP, there is a demand q_i associated with each node $i \in \mathcal{N}$ that must be fulfilled by a single vehicle within a given window $[a_i, b_i]$. The latter means that a vehicle must start service at i at some time between a_i and b_i, which induces additional waiting time if the respective vehicle arrives before the time window. For the depot nodes, we have $q_0 = q_N = 0$, $s_0 = s_N = 0$. Additionally, we assume that the time window of depot nodes overlaps all customer time windows and allows all customers to be served, i.e., $a_0 \leq a_i$, $a_0 \leq b_i - t_{0i}$, $b_0 \geq b_i$ and $b_0 \geq a_i + s_i + t_{iN}$, with $a_N = a_0$ and $b_N = b_0$.

The decision task of the VRPTWRD consists of finding a set of least-cost routes, such that any customer receives their demand from exactly one vehicle within the customer's time window, the capacity restriction of neither vehicle is violated, and each travel leg while a vehicle drives along an arc is assigned a unique driver in $\mathcal{H}$ without overlap. Among all feasible solutions, the VRPTWRD seeks one that minimizes the sum of routing costs c_{ij} for the selected arcs (i, j), where a remote driver teleoperates a vehicle.

We openly admit that our definition of the VRPTWRD captures one particular way to model the different stakeholder aims. We assume that the customers' service requirements are satisfied when delivering them within their time windows. Furthermore, if the logistics provider has a given truck fleet, which cannot be leased on short notice, and a fixedly hired remote driver workforce with long-term contracts, then saving on used trucks and remote drivers barely reduces costs. This leaves the routing costs as our main lever to minimize the costs associated with customer delivery. We leave the multiple alternative ways to model these stakeholder aims to future research.

3.2. A MIP for the VRPTWRD

To formulate VRPTWRD's decision task in a mathematical model, we define the following variables related to the vehicle routes. Let x_{ij} be a binary decision variable that assumes the value of 1 if, and only if, there is a vehicle that traverses arc $(i, j) \in \mathcal{A}$ and, hence, visits node i and then node j, consecutively. Additionally, let u_j be a continuous variable representing the vehicle's cumulative load right after servicing node j.

In addition to vehicle routes, we need to define the schedule of each remote driver. Recall that the number of available drivers H may be smaller than the number of vehicles K. Hence, driving a different vehicle out from the depot node may appear multiple times in a driver's schedule. If not properly addressed, the depot node could appear repeatedly in the sequence, indicating cycles. To overcome this, we create n artificial indices to represent the departures from the depot in the schedule of drivers. Specifically, let $\mathcal{L} = \{N + j : j \in C\} = \{N + 1, \dots, N + n\}$ be the set of artificial indices, such that $N + j$ represents the task related to moving a vehicle from depot node 0 to customer node j. This way, the traversal of arc $(0, j)$ by some vehicle is represented in the driver schedule as task $N+j$, for any $j \in C$. Notably, this is just a modeling strategy to indicate that a vehicle is going from the depot to node j, without limiting the same driver to move further vehicles out from the depot, if necessary. Hence, the indices in $\mathcal{L}$ allow us to effectively represent the driving of different vehicles from the depot by the same driver, at any position in a driver sequence. Note that we do not modify the network, thus the sets $\mathcal{N}$ and $\mathcal{A}$ remain as previously defined. The union $C \cup \mathcal{L}$ can be interpreted as the set of tasks to be performed by the drivers, because each index in this set may be associated with a vehicle moving through arcs of the network.

Example. Fig. 2 shows an example with three vehicle routes and two remote drivers (black and gray). In this example, there are six customer nodes ($n = 6$), thus the depot is represented by indices 0 and 7, and $\mathcal{L} = \{8, \dots, 13\}$. For instance, index 8 represents driving a vehicle from the depot to node 1. We assume that the vehicle capacity Q, typically restricting either the total weight or volume of the loaded shipments, is non-restrictive in this example. Furthermore, we have a service time of $s_i = 2$ and a setup time of $f_i = 1$ at each customer $i = 1, \dots, 6$. Close to each customer node, the figure shows the corresponding time windows; and the arc weights specify the index of the related task and the travel time along this arc. On the right-hand side of this figure, we see the vehicle routes and the driver sequences for the example solution, where the numbers inside parentheses indicate the departure time from the respective position in the vehicle route and driver sequence, respectively. Vehicle 3, for instance, departs from the depot (node 0) at time 2. It must wait for the gray driver, who drives vehicle 1 from the depot to node 5 beforehand. This previous drive takes one time unit plus a setup time of one, so vehicle 3 departs at time 2. After three time units of driving, vehicle 3 reaches customer 6 at time 5, the gray driver is released, and the vehicle must wait to be serviced until the start of the time window at time 6. The black driver, who drives vehicle 3 back to the depot (node 7), is not available before time 15. After three time units of driving, vehicle 3 finally arrives back at the depot at time 18. Based on the assumption of $c_{ij} = t_{ij}$ for all $i, j \in \mathcal{N}$, the total objective value of this solution is 25.

To model the sequence assigned to each driver, we define the binary decision variable y_{ik} that assumes the value of 1 if, and only if, the same driver first operates the vehicle that services node $i \in C \cup \mathcal{L} \cup \{0\}$ and then the vehicle that services node $k \in C \cup \mathcal{L} \cup \{N\}$, consecutively. In this definition, i and k are the tasks related to driving out from i and then out from k. It does not necessarily mean that the same vehicle goes from i to k, as the driver may change vehicles in between.

Example (*cont.*)**.** In our example of Fig. 2, the black driver operates vehicle 1, traveling from node 4 to 2, and then switches to vehicle 3, traveling from node 6 back to depot 7. From the driver' s perspective, this results in $y_{46} = 1$ and $y_{42} = 0$, whereas, from the vehicles' perspective, we have $x_{46} = 0$ and $x_{42} = 1$. In another situation, the black driver stays in vehicle 1 and drives it from node 5 to 4 and from there to 2. In this case, we have $y_{54} = 1$ and $y_{42} = 0$. Hence, while variables x_{ij} model the vehicle sequence, variables y_{ik} model the driver sequence, such that $y_{ik} = 1$ means that one driver operates the vehicle departing from i (to some other node) and then operates the vehicle departing from k (to some other node), consecutively.

Finally, we define one more variable to represent the schedule of the drivers. Let θ_k be a continuous variable equal to the exact time that a driver starts moving the vehicle from node $k \in C \cup \mathcal{L}$ to the next customer in the route. The reader should not confuse this variable with the start of the service at node k, typically considered in VRPTW models. The value of θ_k is the departure time from node k to the next node in the route, and it can only start after the service at node k is completed.

Using the introduced notation, we define the following MIP model for the VRPTWRD, which we denote VRPTWRD-MIP:

$$\min \sum_{(i,j)\in\mathcal{A}} c_{ij}x_{ij} \tag{3.1}$$

$$\text{s.t.} \sum_{j:(i,j)\in\mathcal{A}} x_{ij} = 1, \ \forall i \in C, \tag{3.2}$$

$$\sum_{i:(i,k)\in\mathcal{A}} x_{ik} = \sum_{j:(k,j)\in\mathcal{A}} x_{kj}, \ \forall k \in C, \tag{3.3}$$

$$\sum_{j:(0,j)\in\mathcal{A}} x_{0j} \le K, \tag{3.4}$$

$$u_j \ge u_i + q_j x_{ij} - Q(1 - x_{ij}), \ \forall (i, j) \in \mathcal{A}, \tag{3.5}$$

$$q_i \le u_i \le Q, \forall i \in C, \tag{3.6}$$

$$\sum_{k\in C\cup\mathcal{L}} y_{0k} \le H, \tag{3.7}$$

$$\sum_{\substack{i\in C\cup\mathcal{L}\cup\{0\}:\\ i\neq k}} y_{ik} = \sum_{\substack{j\in C\cup\mathcal{L}\cup\{N\}:\\ j\neq k}} y_{kj}, \ \forall k \in C \cup \mathcal{L}, \tag{3.8}$$

$$\sum_{\substack{k\in C\cup\mathcal{L}\cup\{N\}:\\ k\neq i}} y_{ik} = \sum_{j:(i,j)\in\mathcal{A}} x_{ij}, \ \forall i \in C, \tag{3.9}$$

$$\sum_{\substack{k\in C\cup\mathcal{L}\cup\{N\}:\\ k\neq N+j}} y_{(N+j)k} = x_{0j}, \ \forall j \in C, \tag{3.10}$$

$$\theta_k \ge \theta_i + \sum_{j:(i,j)\in\mathcal{A}} t_{ij}x_{ij} + s_k x_{ik} + f_k(y_{ik} - x_{ik}) - M'_{ik}(1 - y_{ik}), \quad \forall i, k \in C : i \neq k, \tag{3.11}$$

$$\theta_k \ge \theta_i + (t_{ik} + s_k)x_{ik} - M_{ik}(1 - x_{ik}), \ \forall i, k \in C : i \neq k, \tag{3.12}$$

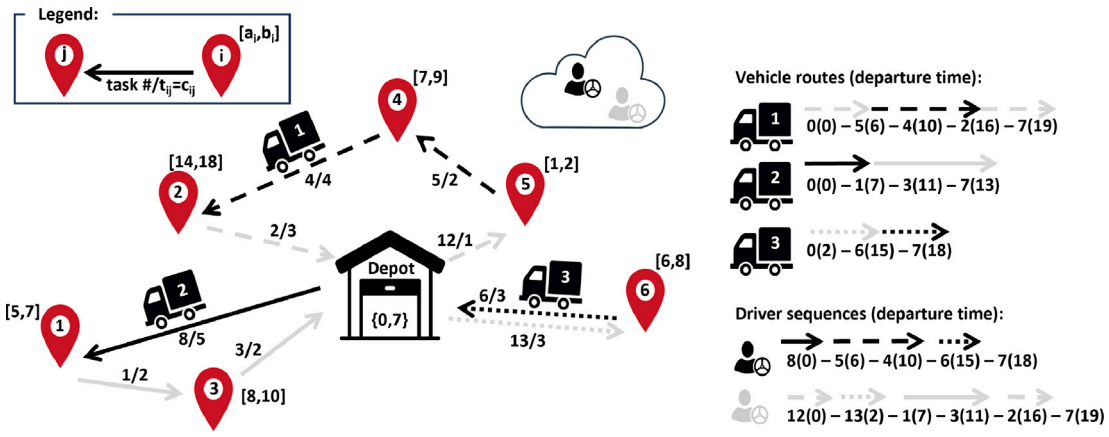


Fig. 2. Example for VRPTWRD.

$$\theta_k \geq \theta_i + \sum_{j:(i,j)\in\mathcal{A}} t_{ij}x_{ij} + f_k y_{ik} - M'_{ik}(1-y_{ik}),\ \forall i \in C, \forall k \in \mathcal{L}, \tag{3.13}$$

$$\theta_k \geq \theta_i + t_{0(i-N)}x_{0(i-N)} + f_k y_{ik} - M'_{ik}(1-y_{ik}),\quad \forall i \in \mathcal{L}, \forall k \in C \cup \mathcal{L} : i \neq k,\ i-N \neq k, \tag{3.14}$$

$$\theta_{(i-N)} \geq \theta_i + (t_{0(i-N)} + s_{(i-N)})x_{0(i-N)},\ \forall i \in \mathcal{L}, \tag{3.15}$$

$$a_k + s_k \leq \theta_k \leq \sum_{j:(k,j)\in\mathcal{A}} (b_j - t_{kj})x_{kj},\ \forall k \in C, \tag{3.16}$$

$$a_0 \leq \theta_k \leq b_{(k-N)} - t_{0(k-N)}x_{0(k-N)},\ \forall k \in \mathcal{L}, \tag{3.17}$$

$$y_{ik} \in \{0,1\},\ \forall i,k \in \mathcal{N} \cup \mathcal{L}, \tag{3.18}$$

$$x_{ij} \in \{0,1\},\ \forall (i,j) \in \mathcal{A}. \tag{3.19}$$

The objective function (3.1) minimizes the total routing costs. Constraints (3.2) and (3.3) ensure that each customer is visited by exactly one vehicle and that each vehicle route starts and ends at the depot, respectively. Constraints (3.4) enforce that the number of routes cannot exceed the number of available vehicles. Constraints (3.5) are the vehicle load propagation constraints which, together with constraints (3.6), ensure that the vehicle capacity is not violated in any route. Constraint (3.7) restricts the maximum number of drivers, and constraints (3.8) ensure the flow conservation in the sequence of tasks of each driver. Constraints (3.9) and (3.10) define the relationship between routes and driver sequences as follows: Constraints (3.9) ensure that the task related to driving the vehicle from customer node i (to any given node j) is assigned to the sequence of a driver. Constraints (3.10) are similar, but consider the departure from the depot node. As mentioned before, the traversal of arc $(0, j)$ in a route is represented by the task $N + j$, to avoid a repetition of index 0 in the sequence of a driver.

Constraints (3.11)–(3.15) control the time propagation and define the schedules of the drivers. They all follow the same reasoning: if the same driver is assigned to tasks i and k consecutively, then executing task k can start only after task i has finished, plus the setup time at k (when applicable). They are divided into four sets only to handle the special cases when using sets C and $\mathcal{L}$, and the modeling technique related to the representation of departures from the depot. Constraints (3.11) consider that both i and k are customer nodes, thus task i finishes at $\theta_i + t_{ij}$, for the index j such that $x_{ij} = 1$. Note that if the visit to k is performed with the same vehicle as the visit of i (i.e., $x_{ik} = 1$), the service time s_k is included in the computation of θ_k, but not the setup time f_k. If the vehicle changes, the setup time is included but not the service time. Constraints (3.12) also consider that i and k are both customer nodes, but they ensure the correct time propagation when both nodes are visited consecutively (i.e., $x_{ik} = 1$) by the same vehicle, but with different drivers (i.e., $y_{ik} = 0$). In this case, constraints (3.11) become inactive and, for this reason, we need constraints (3.12). Constraints (3.13) apply in the case where k is a task related to driving a vehicle out from the depot and, thus, it always includes the setup time, but no service time. Constraints (3.14) and (3.15) consider that i is a task related to driving a vehicle out from the depot and, hence, we need to adjust the travel times appropriately. Valid big-M values for constraints (3.11)–(3.14) are $M_{ik} = \max\{0,\ \max_{j:(i,j)\in\mathcal{A}}\{b_j - t_{ij}\} - (a_k + s_k)\}$ and $M'_{ik} = \max\{0,\ \max_{j:(i,j)\in\mathcal{A}}\{b_j - t_{ij}\} + f_k - (a_k + s_k)\}$. Constraints (3.16) and (3.17) impose the time windows that must be satisfied for the start of the service at each customer. Constraints (3.18) and (3.19) define the domain of the binary decision variables.

3.3. Valid inequalities

To enhance the performance of VRPTWRD-MIP model and our BBC approaches, we introduce a set of valid inequalities. The first inequality explicitly states that the number of remote drivers should not exceed the number of routes:

$$\sum_{k\in C\cup\mathcal{L}} y_{0k} \leq \sum_{j:(0,j)\in\mathcal{A}} x_{0j}. \tag{3.20}$$

The second inequality is based on the reasoning that the total time required to execute all tasks (arc traversals) cannot exceed the available work time. We define the available work time, assuming each driver is available over the entire time horizon (determined by the depot time window). This inequality is defined as follows:

$$\sum_{(i,j)\in\mathcal{A}} t_{ij}x_{ij} \leq (b_0 - a_0)H. \tag{3.21}$$

We also devise a similar inequality considering a reduced time horizon and only the tasks that have to be executed within this horizon, based on the closing time window of a given customer $v \in C$, given as:

$$\sum_{\substack{(i,j)\in\mathcal{A}:\\ b_j - t_{ij} \leq b_v}} (t_{ij} - \max\{0, b_j - b_v\})x_{ij} \leq (b_v - a_0)H. \tag{3.22}$$

Note that this inequality can be further generalized by considering any pair of customers $u, v \in C$ such that $a_u + s_u < b_v$, as follows:

$$\sum_{\substack{(i,j)\in\mathcal{A}:\\ a_i+s_i+t_{ij}\geq a_u+s_u\\ b_j-t_{ij}\leq b_v}} \varphi_{ij}^{uv}x_{ij} \leq (b_v - a_u - s_u)H, \tag{3.23}$$

where φ_{ij}^{uv} represents the minimum amount of time that a driver must necessarily spend moving a vehicle from node i to j within the window $[a_u + s_u, b_v]$ whenever arc (i, j) is traversed. This parameter is computed as:

$$\varphi_{ij}^{uv} = \min\{t_{ij} - \max\{0, b_j - b_v\}, t_{ij} - \max\{0, a_u + s_u - a_i - s_i\}, b_v - a_u - s_u\}. \tag{3.24}$$

Inequalities (3.21) and (3.22) are special cases of (3.23) with $u = 0$ and $v = N$ for the former, and $u = 0$ for the latter. Finally, we rely on existing valid inequalities well-known in the VRPTW literature, which are:

$$\sum_{j:(0,j)\in\mathcal{A}} x_{0j} \geq \left\lceil \frac{1}{Q} \sum_{i\in C} q_i \right\rceil, \tag{3.25}$$

$$x_{ij} = 0,\ \forall (i,j) \in \mathcal{A} : a_i + t_{ij} > b_j \text{ or } q_i + q_j > Q. \tag{3.26}$$

We insert all the presented valid inequalities in the model before invoking the off-the-shelf MIP solver and the BBC methods.

4. Branch-and-Benders-cut methods

Benders decomposition is a highly successful technique in solving challenging MIP models with special structure (Benders, 1962; Rahmaniani et al., 2017). Our VRPTWRD-MIP model has binary variables representing distinct types of decisions interrelated in a certain hierarchy. Specifically, driver variables y_{ik} are related to tasks becoming fully characterized only after vehicle routes are defined according to variables x_{ij}. Hence, we can exploit this hierarchical structure by decomposing each model into a Benders master problem (BMP), involving the vehicle routing decision only. Furthermore, we have a subproblem (SP) determining remote driver schedules, if possible, for the routes defined in the BMP. If no feasible schedule is found for the provided routes, we add a *feasibility* cut to the BMP. This prevents the same output when solving the modified BMP to determine a different set of routes. This type of cut is also known in the literature as *combinatorial* or *no-good* cut (Codato & Fischetti, 2004). The following sections define the BMP, elaborate on two alternative solvers for the SP, and prove the computational complexity of the SP.

4.1. Definition of BMP

Based on the described scheme, we develop a BBC method (Gendron et al., 2016; Moreno et al., 2019), consisting of a tailored branch-and-cut algorithm with feasibility cuts originating from a Benders decomposition scheme and (possibly) combined with other types of valid inequalities. The BBC for the VRPTWRD-MIP is based on the following BMP:

$$\min \sum_{(i,j)\in\mathcal{A}} c_{ij} x_{ij} \tag{4.1}$$

s.t. Constraints (3.2)–(3.6), (3.19),

$$w_j \geq w_i + (s_i + t_{ij}) x_{ij} - M_{ij}(1 - x_{ij}),\ \forall (i,j) \in \mathcal{A}, \tag{4.2}$$

$$a_i \leq w_i \leq b_i,\ \forall i \in \mathcal{N}, \tag{4.3}$$

$$\sum_{(i,j)\in\mathcal{A}(\bar{x})} x_{ij} \leq |\mathcal{A}(\bar{x})| - 1,\ \forall \bar{x} \in \mathcal{X}. \tag{4.4}$$

Here, w_j is a continuous decision variable representing the time the service at node j starts. Constraints (4.2) ensure the correct time propagation between any two consecutive nodes i and j visited in the same route. Additionally, they prevent subtours, since we assume $s_i + t_{ij} > 0$. Valid big-M values for these constraints are $M_{ij} = \max\{0,\ b_i - a_j\}$. Constraints (4.3) enforce respected time windows. Finally, constraints (4.4) are the feasibility cuts guaranteeing that any solution $\bar{x} \in \mathcal{X}$ is forbidden in the BMP. Set $\mathcal{X}$ includes all solutions feasible regarding the route constraints but infeasible regarding remote driver scheduling. In this context, $\mathcal{A}(\bar{x})$ is the subset of arcs traversed in solution $\bar{x}$, i.e., $\mathcal{A}(\bar{x}) = \{(i,j) \in \mathcal{A} : \bar{x}_{ij} = 1\}$. Hence, each inequality in (4.4) cuts off a particular solution $\bar{x}$, by enforcing that at least one arc of this solution cannot be traversed in a feasible BMP solution. Model (4.1)–(4.3) is a well-known formulation for the VRPTW (Desaulniers et al., 2014).

The BBC starts with model (4.1)–(4.3), i.e., without any feasibility cut (4.4). Additionally, before starting the solution process, we add valid inequalities (3.20)–(3.26) to this model. During algorithm execution, every time a candidate incumbent solution $(\bar{x}, \bar{w})$ is found in the search tree, we verify if there is an associated feasible remote driver schedule $(\bar{y}, \bar{\theta})$ for the routes in this solution. Only solutions with an integer $\bar{x}$ can be candidate incumbents. In the following two sections, we define two alternative solvers to verify this for VRPTWRD.

4.2. SP as a MIP

Recall that the SP consists in determining the feasibility of remote driver scheduling for a given set of routes. This section formulates the SP as a MIP model solvable by an off-the-shelf MIP solver, which is given as follows:

$$\min 0 \tag{4.5}$$

s.t. Constraints (3.7), (3.8), (3.18),

$$\sum_{\substack{k\in C\cup\mathcal{L}\cup\{N\}:\\ k\neq i}} y_{ik} = \sum_{j:(i,j)\in\mathcal{A}} \bar{x}_{ij},\ \forall i \in C, \tag{4.6}$$

$$\sum_{\substack{k\in C\cup\mathcal{L}\cup\{N\}:\\ k\neq N+j}} y_{(N+j)k} = \bar{x}_{0j},\ \forall j \in C, \tag{4.7}$$

$$\theta_k \geq \theta_i + t_{i\sigma_i} + s_k \bar{x}_{ik} + f_k(y_{ik} - \bar{x}_{ik}) - M'_{ik}(1 - y_{ik}),\ \forall i,k \in C : i \neq k, \tag{4.8}$$

$$\theta_k \geq \theta_i + t_{ik} + s_k,\ \forall i,k \in C : \bar{x}_{ik} = 1, \tag{4.9}$$

$$\theta_k \geq \theta_i + t_{i\sigma_i} + f_k y_{ik} - M'_{ik}(1 - y_{ik}),\ \forall i \in C, \forall k \in \mathcal{L}, \tag{4.10}$$

$$\theta_k \geq \theta_i + t_{0(i-N)}\bar{x}_{0(i-N)} + f_k y_{ik} - M'_{ik}(1 - y_{ik}),\ \forall i \in \mathcal{L}, \forall k \in C \cup \mathcal{L} : i \neq k,\ i - N \neq k, \tag{4.11}$$

$$\theta_{(i-N)} \geq \theta_i + t_{0(i-N)} + s_{(i-N)},\ \forall i \in \mathcal{L} : \bar{x}_{0(i-N)} = 1, \tag{4.12}$$

$$a_k + s_k \leq \theta_k \leq b_{\sigma_k} - t_{k\sigma_k},\ \forall k \in C, \tag{4.13}$$

$$a_0 \leq \theta_k \leq b_{(k-N)} - t_{0(k-N)},\ \forall k \in \mathcal{L} : \bar{x}_{0(k-N)} = 1. \tag{4.14}$$

The constraints in this model have the same meaning as in (3.9)–(3.17), but they are defined considering a fixed route solution $\bar{x}$. To simplify our notation, σ_i represents the successor of node i on the set of routes, such that $\bar{x}_{ij} = 1$ if, and only if, $j = \sigma_i$. The objective function (4.5) indicates a feasibility problem seeking a feasible set of schedules for the available remote drivers, considering all tasks defined by $\bar{x}$. Note that related customer sequences in the routes impose precedence relations among tasks. Still, the SP is strongly $\mathcal{NP}$-complete which we prove in Section 4.4.

4.3. Solving SP with constraint programming

The SP can also be modeled in an alternative manner, using constraint programming (CP). For the sake of notation, we define the subset $\overline{\mathcal{L}} = \{k \in \mathcal{L} : \bar{x}_{0(k-N)} = 1\}$ given by the tasks related to driving a vehicle out from the depot in the routes of solution $\bar{x}$. Moreover, we define τ_k as the processing time of task k, given by $\tau_k = t_{k\sigma_k}$, for $k \in C$, and $\tau_k = t_{0(k-N)}$, $k \in \overline{\mathcal{L}}$. We also define integer variable θ_k representing the exact time a driver starts performing task k, for each $k \in C \cup \overline{\mathcal{L}}$. Additionally, we define the integer variable d_k specifying the driver that performs task $k \in C \cup \overline{\mathcal{L}}$. The CP model is then formulated as follows:

$$\theta_k + \tau_k + s_{\sigma_k} \leq \theta_{\sigma_k},\ \forall k \in C, \tag{4.15}$$

$$\theta_k + \tau_k + s_{(k-N)} \leq \theta_{\sigma_k},\ \forall k \in \overline{\mathcal{L}}, \tag{4.16}$$

$$(\theta_i + \tau_i + f_k \leq \theta_k) \vee (\theta_k + \tau_k + f_i \leq \theta_i) \vee (d_k < d_i) \vee (d_k > d_i),\ \forall i,k \in C \cup \overline{\mathcal{L}} : i \neq \sigma_k \wedge k \neq \sigma_i, \tag{4.17}$$

$$a_k + s_k \leq \theta_k \leq b_{\sigma_k} - \tau_k,\ \forall k \in C, \tag{4.18}$$

$$a_0 \leq \theta_k \leq b_{(k-N)} - \tau_k,\ \forall k \in \overline{\mathcal{L}}, \tag{4.19}$$

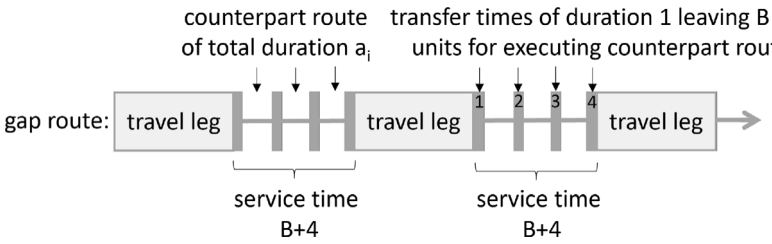


Fig. 3. Schematic structure of driver schedule in the complexity proof.

$$0 \leq d_k \leq H - 1, \ \forall k \in C \cup \overline{\mathcal{L}}. \tag{4.20}$$

If a feasible solution $(\bar{y}, \bar{\theta})$ is found, by solving any of the above models, then $(\bar{x}, \bar{w}, \bar{y}, \bar{\theta})$ is also feasible for the VRPTWRD. Hence, the solution $(\bar{x}, \bar{w})$ is accepted as an incumbent of the BMP. Otherwise, we add $\bar{x}$ to $\mathcal{X}$, resulting in a new feasibility cut.

4.4. Computational complexity of the SP

This section proves that determining the feasibility of remote driver scheduling for a given set of routes (i.e., the SP) remains a challenging computational task. Specifically, we have a set of routes for which the sequence of travel legs (either from a depot to a customer, vice versa, or between two customers) are fixed and given. The remaining decisions of this special case of the VRPTWRD are thus the timing of the travel legs and their assignment to remote drivers. We call this subproblem VRPTWRD'. In the following, we prove that already finding a feasible solution to VRPTWRD' is strongly $\mathcal{NP}$-complete, even if we have only a single remote driver, all transfer times are constant (i.e., we have $f_i = 1$ for all $i \in C$), and time windows are non-restrictive.

The proof is by reduction from the 3-Partition problem (3PP), which is well-known to be strongly $\mathcal{NP}$-complete (Garey & Johnson, 1979). 3PP is defined as follows: Given a set of $3m$ integer values $a_1, a_2, \ldots, a_{3m}$ and a bound B, such that $\frac{B}{4} < a_i < \frac{B}{2}$ for all $i = 1, 2, \ldots, 3m$, is there a partition of $\{1, 2, \ldots, 3m\}$ into m disjoint sets $A_1, A_2, \ldots, A_m$, such that $\sum_{i \in A_k} a_i = B$ for all $k = 1, 2, \ldots, m$?

For an arbitrarily given instance of 3PP, we construct an instance of the VRPTWRD' as follows. We have one route, the *gap route*, which is exceptionally long. Its $m + 1$ travel legs plus m service intervals at customer locations, each with a service time of $B + 4$ time units, span the whole planning horizon. Furthermore, we introduce $3m$ additional tours, which we call the *counterpart tours*, visiting merely one customer each. The travel legs back and forth their respective customer take the counterpart tours $\frac{a_i}{2}$ travel time, and their service times are $s_i = 0$. All customers involved in the gap route and the counterpart routes have non-restrictive time windows. The question we ask is whether a single remote driver can feasibly execute all routes within the planning horizon.

To achieve this, all counterpart routes must fit into the service time intervals of the gap route. Since we have transfer time $f_i = 1$ for all $i \in C$ and the total driving time of all counterpart routes is $m \times B$, each service time interval offers driving time for exactly three counterpart tours. Note that the restricted integer vales a_i of 3PP (i.e., $\frac{B}{4} < a_i < \frac{B}{2}$) enforce that it takes exactly the sum of three integers to equate bound B. Hence, in VRPTWRD', each service time of duration $B + 4$, also leaves a gap for exactly three counterpart tours with a total driving time of B. The four additional service time units are needed for four transfers, each of duration 1 (i.e., the transfer time denoted with 1 in Fig. 3 for switching from the gap route to the first counterpart route, transfer times 2 and 3 for switching among counterpart routes, and transfer time 4 for switching from the final counterpart route back to the gap route). This induces that the travel legs of each counterpart route back and forth its customer must be executed in direct succession without intermediate switching. The necessary additional transfer time would not fit into the available service time intervals. Therefore, each resulting counterpart tour has a total duration of a_i, equal to its integer counterpart of 3PP. Thus, exactly three counterpart routes, whose total durations sum up to B, must be assigned to each service time interval of the gap route and the one-to-one mapping between both problems is readily available. The resulting driver schedule is schematically depicted in Fig. 3.

5. Computational study

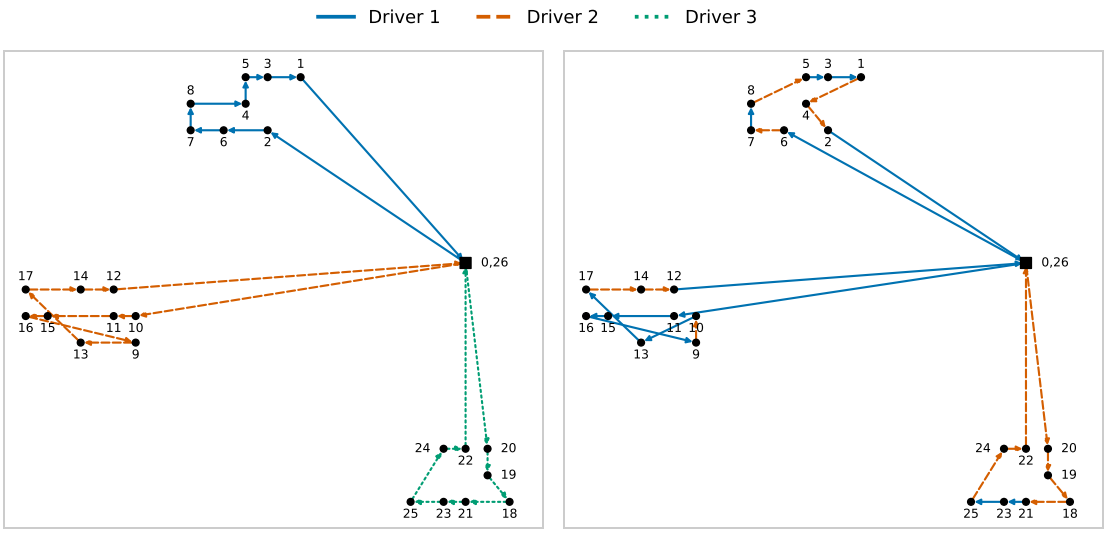
This section describes the results of our computational experiments with the proposed compact VRPTWRD-MIP model and our BBC methods. All approaches were implemented in C++ using the Concert library of IBM CPLEX Optimization Studio v.22.1.1, relying on this software's general-purpose MIP solver and callback functions. We use the default settings of CPLEX and set the time limit to 3600 s for each instance and solution approach. All experiments reported in this section were executed on a Linux PC with Intel Xeon E5-2680 2.7 GHz processor (16 cores) and 192 GB of memory.

We use the VRPTW benchmark instances proposed by Solomon (1987). There are 56 instances, divided into 6 groups named C1, R1, RC1, C2, R2, and RC2. Instances in groups named with letters C, R, and RC have customer positions that are clustered, randomly distributed, and mixed of clustered and randomly distributed customers. Instances in groups with a suffix 1 have narrower time windows and/or lower vehicle capacities than instances in groups with a suffix 2. Group C1 has nine instances, group R1 has twelve instances, groups RC1, C2, and RC2 have eight instances each. Finally, group R2 has eleven instances. We assume Euclidean distances, truncated after the first decimal place, as is the common practice in the literature. Travel times are defined as equal to the computed distances. Service times are set as defined on the instances (90 for instances in C1 and C2, and 10 for instances in the remaining classes). All instances are originally defined with 100 customers. In the default case, we perform experiments considering reduced versions of these instances with the first 25 customers only. In Section 5.2.3, we also consider larger instances with 50 and 100 customers.

5.1. Illustrative example

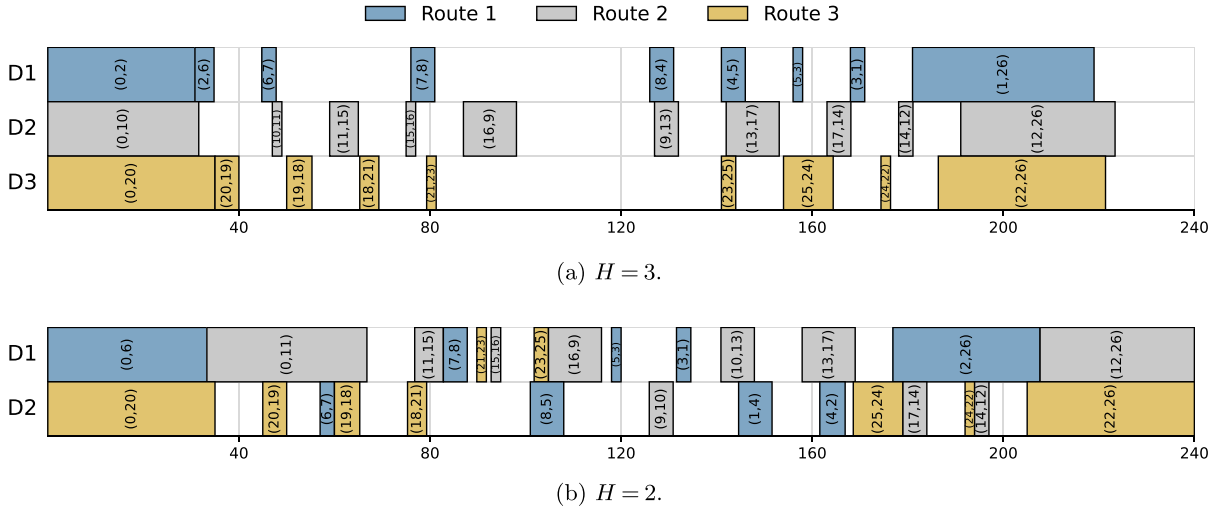
Before analyzing the computational performance of the proposed models and solution methods, we illustrate the results obtained for a representative benchmark instance that was arbitrarily selected, namely RC104 with 25 customers. The purpose is to highlight the relevance of incorporating remote driver assignment with routing decisions.

In the selected instance, the vehicle capacity is equal to 200, customer demands belong to set $\{10, 20, 30, 40\}$, service times are equal to 10 for all customers, time windows vary inside the interval $[0, 199]$, and the depot time window is $[0, 240]$. The setup time f_i is zero. When ignoring remote driver assignment decisions, we obtained an optimal VRPTW solution that has an objective value of 306.6, corresponding to three routes $(0, 2, 6, 7, 8, 4, 5, 3, 1, 26)$, $(0, 10, 11, 15, 16, 9, 13, 17, 14, 12, 26)$, and $(0, 20, 19, 18, 21, 23, 25, 24, 22, 26)$. Recall that 0 and 26 are the depot nodes.



(a) $H = 3$. (b) $H = 2$.

Fig. 4. Optimal vehicle routes for instance RC104 when enforcing $H = 3$ and $H = 2$ drivers. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) $H = 3$.

(b) $H = 2$.

Fig. 5. Optimal driver schedules for instance RC104 when enforcing $H = 3$ and $H = 2$ drivers. Each bar labeled (i, j) denotes a driving task that moves a vehicle from node i to node j. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 4 depicts the optimal vehicle routes for two scenarios with a given workforce of either $H = 3$ or $H = 2$ drivers. Each route is represented in the two-dimensional plane using a visual identity in which arcs are directed and colored according to the driver performing the corresponding driving task. For $H = 3$ (Fig. 4(a)), each driver is assigned to exactly one route, and the resulting solution coincides with the optimal VRPTW solution, with an objective value of 306.6.

When the workforce is reduced to $H = 2$ drivers (Fig. 4(b)), three routes are still obtained, and each route serves the same set of customers. However, the visiting sequences are slightly modified to ensure feasibility with fewer drivers. In particular, the first route changes from $(0, 2, 6, 7, 8, 4, 5, 3, 1, 26)$ to $(0, 6, 7, 8, 5, 3, 1, \mathbf{4}, \mathbf{2}, 26)$, where customers 4 and 2 are moved to the end of the route. Moving customer 2 from the first to the last position is possible because the respective time window (i.e., $[0, 191]$) is wide enough. The second route also changes marginally, becoming $(0, 11, 15, 16, 9, \mathbf{10}, 13, 17, 14, 12, 26)$, while the third route remains unchanged. The optimal objective value in this case is 314.1. Hence, a workforce reduction of 33.3% results in an increase in routing costs of less than 2.5%. A further reduction of the workforce is not possible, as no feasible solution exists for $H = 1$. Fig. 4(b) also illustrates how driver assignments change when the workforce is reduced to $H = 2$. In this case, drivers are no longer associated with a single vehicle route, but instead alternate between routes over time. This reassignment of driver legs is visually highlighted by the change in the arc representation (color and line style) along the routes.

Fig. 5 presents the corresponding driver schedules for the two workforce scenarios. The horizontal axis explicitly represents time and is identical in both subfigures, allowing a direct comparison of makespan. Each box labeled (i, j) corresponds to a driving task that moves a vehicle from node i to node j. In each route, we use the same color to represent all tasks in this route. For $H = 3$, each driver executes tasks belonging to a single route, resulting in schedules with substantial idle times. For $H = 2$, drivers execute tasks from multiple routes, clearly illustrating how route segments are interleaved across drivers. This visual comparison highlights the reduction in driver idleness when the workforce is reduced, which enables the consolidation of driving tasks across routes.

Moreover, Fig. 5 allows us to quantify the utilization of the remote drivers through their idle times. For $H = 3$, although routing costs are minimal, a substantial amount of driver capacity remains unused: the three drivers exhibit idle times of approximately 144.2, 130.9, and 138.3 time units, respectively, resulting in a total idle time of 413.4 over the planning horizon. Given a horizon of 240 time units per driver, this corresponds to an average idle rate of about 57.4% of the available driver time. In contrast, when the workforce is reduced to $H = 2$, we observe a significant improvement in driver utilization. The two drivers present idle times of 57.9 and 108.0, yielding a total idle time of 165.9. This represents an average idle rate of 34.6% and a reduction of around 60% in total idle time relative to the case with three drivers. This significant decrease in idle time is achieved by allowing drivers to alternate between different vehicle routes, as illustrated in Figs. 4(b) and 5(b). From a managerial perspective, these results highlight that a modest increase in routing cost may enable significant workforce efficiency gains, suggesting that remote driving can be an effective lever to reduce labor underutilization.

5.2. Computational performance

We now analyze the computational performance of the proposed solution approaches for different configuration settings and scenarios. We consider three approaches: solving the compact VRPTWRD-MIP model, consisting of (3.1)–(3.19), using CPLEX as a general-purpose MIP solver; and two variants of our BBC method. In the variant named BBC-MIP, feasibility cuts are separated using the MIP model (4.5)–(4.14), while in the variant denoted BBC-CP the CP model (4.15)–(4.20) is used for solving SP. Preliminary results have shown a negligible performance impact of reasonable values of setup times. Hence, we set all setup times to zero. Note that the online appendix provides further performance results, detailing the number of nodes and cuts as well as the impact of specific instance data.

5.2.1. Comparison of the three approaches

Table 1 presents a summary of the results obtained with the three considered approaches, all including the valid inequalities presented in Section 3.3. For each maximum number of remote drivers (H) and instance group, the table lists the following performance criteria: the average upper bound (UB), computed as the average solution value obtained for the instances in the group; the average gap (Gap) in %, considering only instances for which a feasible solution has been found, where the gap in each instance is the relative difference between the lower and upper bounds; the average computation time (Time) in seconds for all processed instances in the group; the number of instances solved to proven optimality (O); the number of instances in which the approach found a feasible solution but could not prove its optimality (F); the number of instances in which the approach terminated without finding any feasible solution (NF); and the number of instances that were proven infeasible by the approach (IN). The line *All*, presented for each value of H, shows the aggregated results considering all instances. Here, we highlight the best result among the three approaches in columns Gap, Time, and O in boldface. The average UB values should be interpreted cautiously whenever column NF has a positive value. In these cases, averages of each approach may refer to different instances. It is worth mentioning that the optimal solution of neither instance has more than 8 routes. Hence, the instances with $H = 10$ have a non-restrictive remote driver workforce.

The results presented in Table 1 indicate that the three approaches are successful in solving the VRPTWRD instances. For a non-restrictive driver workforce of $H = 10$, they determine feasible solutions to all instances, most of which are solved to proven optimality. Instances in group RC2 are the most difficult for this choice of H, as none of the approaches could prove optimality for three of these instances within the time limit. When H decreases, the overall computational performance of the approaches is negatively affected by the more restrictive driver schedules. Despite this general trend, the deterioration of algorithmic performance (i.e., both related to solution time and quality) is not monotonic in H. Hence, the tightness of the driver schedule is not the only complexity driver (see Online appendix B). Instances turn out infeasible only for workforce sizes less than or equal to $H = 2$. Specifically, six instances are infeasible for $H = 2$ and an additional 14 instances are infeasible for $H = 1$, all from classes R1 and RC1. All three solution approaches detect these cases quickly.

In all instance groups and values of H, the BBC approaches consistently outperform the compact model solved with the MIP solver. For less restrictive values of H, the computational performance of the three approaches is on a similar level in terms of computation times and number of proven optimal solutions. However, for H equal to 1 and 2, we observe that the performance of the MIP solver with the compact model degrades significantly, and the BBC approaches excel. For instance, BBC-CP achieves 44 optimal solutions for $H = 2$ and 33 for $H = 1$, while the compact model achieves only 35 and 26, respectively. The average solution times show even more pronounced differences: for $H = 2$, BBC-CP requires 486.3 s versus 1002.2 s for the compact model (51% reduction); for $H = 1$, the reduction is 71% (202.7 vs. 694.9 s). The two BBC methods deliver comparable overall performance, with BBC-CP holding a slight edge. For most values of H, this method result in the best average computation time, average gap, and number of instances solved to proven optimality. Based on these results, we focus our analysis on the results of the BBC approaches in the remainder of this section.

5.2.2. Influence of the valid inequalities

We now analyze how the valid inequalities proposed in Section 3.3 influence the performance of the BBC approaches. We focus our analysis on inequalities (3.21)–(3.23), since they are tailored for the VRPTWRD. Inequalities (3.25) and (3.26) are well-known and widely adopted in the VRPTW literature.

Tables 2 and 3 summarize the results obtained for BBC-MIP and BBC-CP, respectively, in experiments turning off one family of valid inequalities at a time. For each configuration, the tables show the percentage variations for the average upper bound (UB), average gap (Gap), and average computation time (Time). Furthermore, we report the absolute variation of the number of optimal (O) and feasible (F) solutions, as well as the number of instances with no feasible solution found (NF). All variations are computed using the results of the corresponding BBC method in Table 1 as a baseline. Hence, the values in Tables 2 and 3 show the methods' performance changes related to their execution with all inequalities; negative values in columns UB, Gap, Time, and NF indicate a performance improvement when excluding valid inequalities, while negative values in column O identify the opposite.

The results in Tables 2 and 3 clearly indicate the importance of valid inequalities (3.21) to the performance of both BBC-MIP and BBC-CP. For all restrictive values of H (i.e., $H \leq 5$), we observe significant increases in the gap and computation time, as well as a significant reduction in the number of instances with a proven optimal solution. Without these valid inequalities, the computation times of the BBC approaches increase by a factor of around 4 for some values of H. We also observe significant increases in the gap values. One important remark is that the reductions in the UB should be interpreted cautiously because they do not correspond to the same instances. In fact, the BBC version without valid inequalities (3.21) could not find a feasible solution for many instances (see column NF), and the average UB is computed considering instances with feasible solutions only.

The impact of valid inequalities (3.22) and (3.23) is less evident. When turning them off, we observe reductions in the computation times for both BBC-MIP and BBC-CP, but the upper bounds and gaps mostly increase, especially for BBC-CP. For instance, by turning off (3.22) in BBC-CP, we observe a time reduction of around 15% on average for $H = 3$, but the gap increases by approximately 31%. Still, the method

Table 1
Performance results of the compact model and its two BBC competitors, whose subproblems either solve the MIP or CP model, for workforces with at most H remote drivers.

H	Group	Compact model							BBC-MIP							BBC-CP						
		UB	Gap	Time	O	F	NF	IN	UB	Gap	Time	O	F	NF	IN	UB	Gap	Time	O	F	NF	IN
10	C1	190.6	0.0	6.0	9	0	0	0	190.6	0.0	0.1	9	0	0	0	190.6	0.0	0.1	9	0	0	0
	R1	463.4	0.0	360.9	11	1	0	0	463.4	0.0	127.8	12	0	0	0	463.4	0.0	143.9	12	0	0	0
	RC1	350.2	0.0	29.5	8	0	0	0	350.2	0.0	1.9	8	0	0	0	350.2	0.0	2.1	8	0	0	0
	C2	214.5	0.0	25.0	8	0	0	0	214.5	0.0	0.7	8	0	0	0	214.5	0.0	0.7	8	0	0	0
	R2	382.1	0.0	141.4	11	0	0	0	382.1	0.0	15.2	11	0	0	0	382.1	0.0	16.1	11	0	0	0
	RC2	319.4	12.3	1775.3	5	3	0	0	319.3	4.5	1368.8	5	3	0	0	319.3	4.1	1363.1	5	3	0	0
	All	331.3	1.8	367.5	52	4	0	0	331.3	**0.6**	**226.3**	**53**	3	0	0	331.3	**0.6**	229.2	**53**	3	0	0
5	C1	190.6	0.0	4.8	9	0	0	0	190.6	0.0	0.1	9	0	0	0	190.6	0.0	0.1	9	0	0	0
	R1	463.4	0.0	288.9	11	1	0	0	465.1	0.0	532.4	11	0	1	0	463.4	0.0	211.9	12	0	0	0
	RC1	350.2	0.0	32.6	8	0	0	0	350.2	0.0	3.3	8	0	0	0	350.2	0.0	2.3	8	0	0	0
	C2	214.5	0.0	45.5	8	0	0	0	214.5	0.0	0.7	8	0	0	0	214.5	0.0	0.8	8	0	0	0
	R2	382.1	0.0	181.6	11	0	0	0	382.1	0.0	13.5	11	0	0	0	382.1	0.0	12.9	11	0	0	0
	RC2	319.8	13.0	1840.7	5	3	0	0	319.3	4.3	1376.9	5	3	0	0	319.3	4.1	1374.1	5	3	0	0
	All	331.3	1.9	372.5	52	4	0	0	329.2	**0.6**	314.0	52	3	1	0	331.3	**0.6**	**244.7**	**53**	3	0	0
3	C1	190.6	0.0	8.9	9	0	0	0	190.6	0.0	0.2	9	0	0	0	190.6	0.0	0.2	9	0	0	0
	R1	470.0	0.5	736.0	9	1	2	0	463.4	0.0	231.1	12	0	0	0	463.4	0.0	205.1	12	0	0	0
	RC1	350.2	0.0	35.5	8	0	0	0	350.9	0.0	452.6	7	0	1	0	350.9	0.0	452.5	7	0	1	0
	C2	214.5	0.0	50.6	8	0	0	0	214.5	0.0	1.1	8	0	0	0	214.5	0.0	0.9	8	0	0	0
	R2	382.1	0.0	173.5	11	0	0	0	382.1	0.0	13.4	11	0	0	0	382.1	0.0	13.5	11	0	0	0
	RC2	319.4	13.1	1853.8	5	3	0	0	319.3	4.4	1370.6	5	3	0	0	319.3	3.9	1370.9	5	3	0	0
	All	330.2	2.0	470.4	50	4	2	0	331.0	**0.6**	312.8	**52**	3	1	0	331.0	**0.6**	**307.2**	**52**	3	1	0
2	C1	190.6	0.0	13.1	9	0	0	0	190.6	0.0	0.3	9	0	0	0	190.6	0.0	0.3	9	0	0	0
	R1	435.1	0.0	1550.5	1	0	6	5	422.1	0.5	503.3	6	1	0	5	422.1	0.2	463.9	6	1	0	5
	RC1	325.6	0.7	2403.3	2	0	5	1	332.2	0.8	907.1	5	1	1	1	332.2	0.9	914.4	5	1	1	1
	C2	214.5	0.0	36.6	8	0	0	0	214.5	0.0	0.9	8	0	0	0	214.5	0.0	0.9	8	0	0	0
	R2	382.1	0.0	272.1	10	1	0	0	383.2	0.0	16.2	11	0	0	0	382.1	0.0	14.9	11	0	0	0
	RC2	322.5	13.5	1860.6	5	3	0	0	319.3	4.1	1369.9	5	3	0	0	319.3	4.2	1772.2	5	3	0	0
	All	294.2	2.7	1002.2	35	4	11	6	309.2	**0.8**	**436.5**	**44**	5	1	6	308.9	**0.8**	486.3	**44**	5	1	6
1	C1	196.2	0.0	75.2	9	0	0	0	196.9	0.0	4.3	9	0	0	0	196.2	0.0	0.5	9	0	0	0
	R1	–	–	0.7	0	0	0	12	–	–	0.4	0	0	0	12	–	–	0.3	0	0	0	12
	RC1	–	–	0.8	0	0	0	8	–	–	0.4	0	0	0	8	–	–	0.3	0	0	0	8
	C2	214.5	0.0	109.8	7	1	0	0	214.5	0.0	1.1	8	0	0	0	214.5	0.0	1.0	8	0	0	0
	R2	385.2	1.8	1666.0	6	4	1	0	385.3	0.0	340.1	10	0	1	0	382.1	0.0	20.7	11	0	0	0
	RC2	323.7	17.5	2377.3	4	2	2	0	319.3	5.0	1386.4	5	3	0	0	319.3	4.5	1387.9	5	3	0	0
	All	286.3	4.5	694.9	26	7	3	20	282.7	1.1	265.8	32	3	1	20	284.4	**1.0**	**202.7**	**33**	3	0	20

Table 2
Variations on the performance of BBC-MIP when turning off each family of valid inequalities, for different choices of H remote drivers. The results in Table 1 are used as baseline for the variations.

H	BBC-MIP without (3.21)						BBC-MIP without (3.22)						BBC-MIP without (3.23)					
	UB	Gap	Time	O	F	NF	UB	Gap	Time	O	F	NF	UB	Gap	Time	O	F	NF
10	0.01%	−8.81%	1.21%	0	0	0	0.00%	−15.06%	1.36%	0	0	0	0.00%	−0.47%	1.86%	0	0	0
5	−12.57%	50.50%	352.86%	−18	0	18	0.64%	−9.44%	−22.27%	1	0	−1	0.11%	−3.33%	−0.37%	0	0	0
3	−1.19%	2.12%	39.66%	−2	0	2	0.08%	−10.46%	−18.13%	1	0	−1	0.09%	−6.03%	−22.67%	1	0	−1
2	−5.41%	27.61%	132.26%	−5	−1	9	−0.08%	−0.83%	−2.67%	0	0	0	−0.18%	5.18%	9.19%	0	−1	1
1	−16.43%	3.94%	290.46%	−12	−1	13	−2.03%	−1.04%	52.94%	−2	0	2	−0.18%	−12.47%	1.34%	0	0	0

Table 3
Variations on the performance of BBC-CP when turning off each family of valid inequalities, for different choices of H remote drivers. The results in Table 1 are used as baseline for the variations.

H	BBC-CP without (3.21)						BBC-CP without (3.22)						BBC-CP without (3.23)					
	UB	Gap	Time	O	F	NF	UB	Gap	Time	O	F	NF	UB	Gap	Time	O	F	NF
10	0,01%	0,67%	1,51%	0	0	0	0,00%	−6,81%	0,83%	0	0	0	0,00%	2,30%	−0,34%	0	0	0
5	−13,11%	57,38%	480,57%	−19	0	19	0,01%	8,21%	4,55%	0	0	0	−0,51%	8,20%	23,64%	−1	0	1
3	−1,19%	9,64%	47,17%	−2	0	2	0,08%	31,08%	−15,32%	1	0	−1	0,08%	8,98%	−20,83%	1	0	−1
2	−5,33%	40,69%	139,45%	−6	0	8	0,00%	0,67%	−10,90%	0	0	0	0,22%	−38,25%	16,70%	−1	−3	4
1	−16,94%	19,74%	412,05%	−13	−1	14	0,06%	8,05%	−0,75%	0	0	0	0,00%	1,00%	−1,47%	0	0	0

without (3.22) could prove optimality for one more instance. The benefit of (3.22) for BBC-MIP is also unclear, depending on the choice of H. With $H = 5$ and 3, turning off (3.22) is beneficial for BBC-MIP, while for $H = 1$ the performance of the approach degrades regarding average computation time and number of proven optimal solutions. The influence of valid inequalities (3.23) seems generally (mostly) positive. Except for $H = 3$, when turning them off, the number of proven optimal solutions remains the same or lowers, and the average gap and computation times increase for several values of H.

To conclude, our tests highly recommend the inclusion of valid inequalities (3.21) for both BBC-MIP and BBC-CP. Instead, valid inequalities (3.22) should rather be excluded if the focus is on reducing

Table 4
Average results of BBC-MIP on instances with 50 and 100 customers.

n	H	Group	UB	Gap	Time	T_{Sub}	T_{Best}	Nodes	Cand	Feas	Cap	R	D	O	F	NF	IN
50	10	C1	361.7	0.0	3.5	1.1	3.2	89.2	20.2	10.8	247.2	5.0	5.0	9	0	0	0
		R1	758.2	12.4	3009.5	903.2	2164.7	62 737.4	66.3	52.8	53 360.8	8.0	7.6	2	7	3	0
		RC1	742.6	14.2	2757.7	0.4	2631.6	213 600.3	200.5	177.8	69 439.0	6.6	6.6	2	6	0	0
		C2	357.5	0.0	3.0	0.6	2.5	453.9	30.5	20.6	105.0	2.8	2.8	8	0	0	0
		R2	618.7	6.8	2804.5	1.3	1513.2	466 163.7	140.5	111.3	56 020.5	4.0	4.0	3	8	0	0
		RC2	575.7	12.8	2447.9	0.9	1429.7	619 484.5	103.1	79.4	26 705.1	4.4	4.4	3	5	0	0
		All	571.5	7.6	1940.4	194.2	1295.6	224 102.9	92.8	74.6	36 228.1	5.1	5.0	27	26	3	0
	5	C1	361.7	0.0	4.8	2.5	4.5	129.7	35.1	25.4	217.3	5.0	5.0	9	0	0	0
		R1	733.9	12.3	3305.9	621.5	2315.9	126 390.6	279.8	263.0	60 545.4	7.5	5.0	1	9	2	0
		RC1	778.3	16.2	3600.0	1267.5	2140.2	171 347.1	4636.1	4623.1	60 205.5	7.3	5.0	0	6	2	0
		C2	357.5	0.0	3.6	0.9	3.1	481.3	43.3	34.3	102.9	2.8	2.8	8	0	0	0
		R2	619.8	7.0	2755.9	3.9	1665.9	462 608.6	70.5	42.8	55 523.2	4.0	3.9	3	8	0	0
		RC2	576.1	12.9	2530.5	3.7	1134.1	647 702.8	62.4	37.0	32 851.5	4.4	4.4	3	5	0	0
		All	568.3	7.7	2127.5	316.1	1220.4	235 050.0	756.8	739.5	37 223.8	5.1	4.3	24	28	4	0
100	15	C1	826.7	0.0	220.6	13.7	159.5	1296.7	23.1	8.2	4787.6	10.0	10.0	9	0	0	0
		R1	1318.4	24.8	3379.5	14.5	2842.3	23 384.7	233.8	219.7	51 601.1	15.8	14.4	1	11	0	0
		RC1	1551.8	29.3	3600.0	19.3	3243.1	37 975.9	376.6	361.8	60 899.6	16.0	14.9	0	8	0	0
		C2	587.4	0.0	37.7	4.6	35.9	4448.5	28.9	14.5	697.3	3.0	3.0	8	0	0	0
		R2	932.0	17.3	3289.0	9.5	3188.0	108 657.2	147.8	110.3	57 879.8	6.3	6.3	1	10	0	0
		RC2	1071.3	20.8	3171.8	10.8	3078.5	104 865.8	170.3	130.4	38 399.4	7.8	7.8	1	7	0	0
		All	1057.1	15.9	2393.2	12.1	2169.1	47 604.2	165.1	142.4	37 481.2	10.1	9.6	20	36	0	0
	10	C1	826.7	0.0	230.8	24.1	171.0	1252.8	29.8	14.9	4929.1	10.0	10.0	9	0	0	0
		R1	1308.3	24.4	3401.6	22.2	2325.9	23 724.6	196.6	182.6	51 918.9	15.9	10.0	1	11	0	0
		RC1	1523.8	28.1	3600.0	265.4	3333.5	42 272.1	280.0	260.8	58 351.4	15.4	10.0	0	8	0	0
		C2	587.4	0.0	32.6	8.9	31.8	1587.9	25.8	13.5	339.9	3.0	3.0	8	0	0	0
		R2	936.5	17.7	3289.7	23.7	3198.4	108 255.3	132.8	93.5	57 769.0	6.7	6.7	1	10	0	0
		RC2	1062.8	20.4	3189.3	41.2	3119.3	106 356.0	132.8	93.4	40 100.4	7.3	7.3	1	7	0	0
		All	1050.6	15.6	2398.4	58.3	2080.5	48 009.0	135.6	112.4	37 378.2	10.0	8.0	20	36	0	0

computation times, except for BBC-MIP and restrictive values of H. The use of (3.23) seems generally advisable, especially for BBC-CP.

5.2.3. Performance on larger instances

We further verify the performance of the BBC algorithm on instances with 50 and 100 customers. The ones with 100 customers are the original Solomon instances, and those with 50 customers consider only the first 50 entries in each instance. Table 4 shows the results obtained with BBC-MIP on both sets of instances, with different choices of H. In these experiments, we execute the method with all valid inequalities. For instances with 50 customers, we set $H = 10$ and 5. The former is non-restrictive for most instances, whereas the latter is restrictive for around half of the instances, without making any of them infeasible. Analogously, we select $H = 15$ and 10 for instances with 100 customers. In addition to the same metrics reported in previous tables, the columns in Table 4 show the total time spent on solving the Benders subproblem (T_{Sub}), the elapsed time when the best solution was found (T_{Best}), the total number of nodes (Nodes), the number of candidate incumbents verified in the Benders subproblem (Candidates), the number of feasibility cuts generated by calling the Benders subproblem (Feas Cuts), the number of rounded capacity inequalities generated by CVRPSep (Cap Cuts), and the numbers of routes (R) and drivers (D) in the final solution. All these values are averages over all instances (with at least a feasible solution) in the corresponding group.

The results in Table 4 show that the BBC approach is effective even on larger instances, finding feasible solutions to almost all 50-customer instances, and all 100-customer instances. The only exceptions are three instances in group R1 with $H = 10$, and two instances each in groups R1 and RC1 with $H = 5$. Moreover, we obtain proven optimal solutions for 20 of the 100-customer instances for both H values, and for 24 and 27 of the 50-customer instances for $H = 10$ and 5, respectively. All instances in groups C1 and C2 are solved to optimality in all configurations. Conversely, the resulting average gaps for instances in groups R1, RC1, and RC2 are large, showing that these instances are challenging for BBC-MIP. Most of the computation time of the method is spent on solving the BMP.

5.3. Managerial insights

We now analyze the relevance and potential benefits of integrating remote driver scheduling and vehicle routing, considering the number of drivers, costs, idleness, and operational flexibility.

5.3.1. Number of routes and workforce size

Fig. 6 illustrates the relationship between the number of drivers and the number of routes across the six instance groups for $H \in \{1, 2, 3, 5, 10\}$, considering all instances solved to proven optimality with BBC-MIP. We observe a similar behavior across all instance groups regarding workforce reduction (i.e., decreasing H): the average number of routes remains essentially stable, showing that remote driving enables significant workforce reductions without requiring additional vehicle routes in most scenarios. Except for the smallest workforce sizes, the route number remains virtually unchanged in all instance groups. When the workforce is reduced to a single driver, we observe distinct patterns. In clustered instances (groups C1 and C2) it still possible to find feasible schedules, but the average number of routes increases slightly to accommodate the constrained driver availability. In groups R1 and RC1, no feasible solutions could be found for $H = 1$, because of the tight time windows and extended geographical dispersion within these instance groups. Instances in groups R2 and RC2 remain feasible for $H = 1$, reflecting their wider time windows and extended flexibility to switch the driver among multiple vehicles while satisfying all service requirements.

Table 5 presents the relationship between the number of routes and the workforce size of remote drivers in the best solutions obtained for the benchmark instances. All values were obtained from the results summarized in Table 1 considering $H \in \{1, 2, 3, 5, 10\}$. Entries in the main diagonal of this table indicate the number of instances, in which the number of drivers is identical to the number of routes (which is always feasible for any instance). Entries above the main diagonal correspond to instances, in which we could further reduce the number of drivers. Note that an instance may be counted in more than one entry, as it may allow solutions with different numbers of drivers.

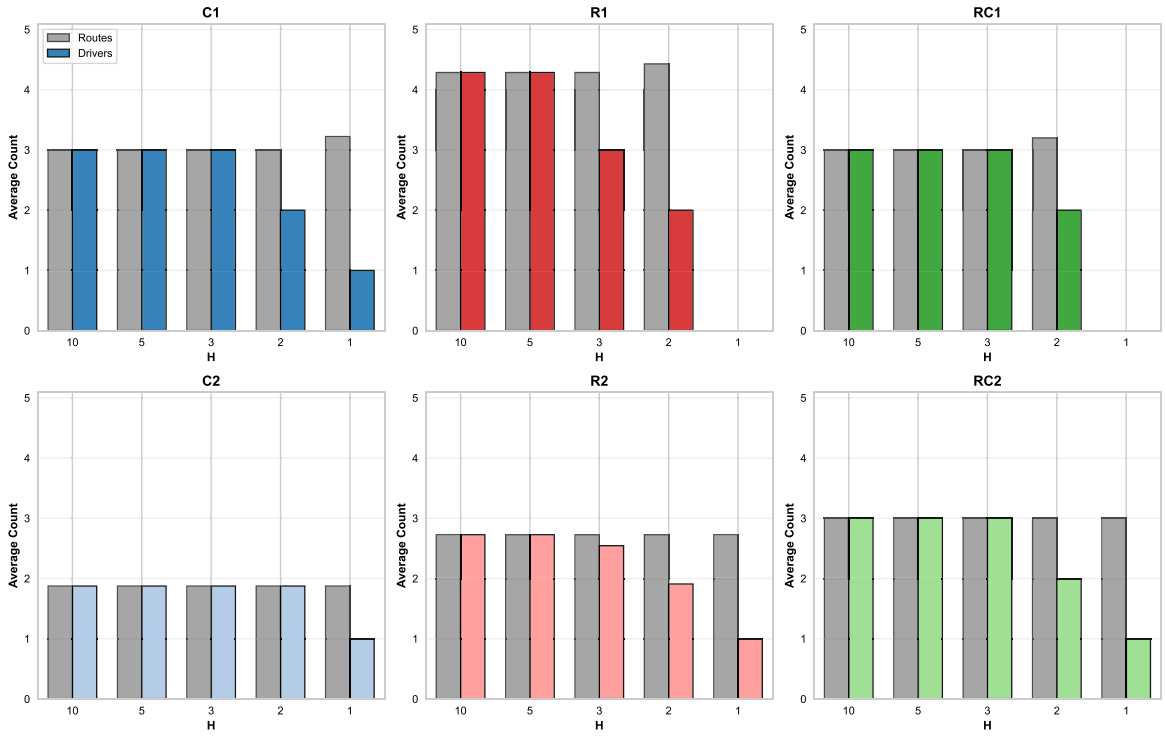


Fig. 6. Average number of routes and drivers as a function of workforce size H across the six instance groups.

Table 5
Number of instances with a solution having the corresponding number of routes and remote drivers, for $H \in \{1, 2, 3, 5, 10\}$.

# Routes	# Remote drivers							
	8	7	6	5	4	3	2	1
8	1	–	–	1	1	1	–	–
7		1	–	1	1	1	–	–
6			1	–	1	1	–	–
5				4	4	4	3	–
4					9	9	7	1
3						27	25	18
2							11	11
1								2

Moreover, since we solve the instances using only the values of H specified above, there are some empty entries. However, if an instance is feasible with a given number of remote drivers, it still is with more drivers.

The results in Table 5 again indicate the significant potential to reduce the workforce of remote drivers. In one instance, namely R101, we obtain a solution with 8 routes and 3 drivers, which is a large difference from the trivial allocation of one driver per route. Among the 56 instances, it was possible to reduce one driver in 49 cases, two drivers in 31 instances, three drivers in seven cases, and four drivers in two instances. We summarize these results as follows:

Finding 1. Remote driving promises a significant workforce reduction compared to traditional on-board driving. Furthermore, reducing the workforce barely deteriorates the vehicle routes, whose costs are mostly unaffected or merely increase by a small single-digit percentage. Hence, remote driving is a promising tool for facing future driver shortages.

5.3.2. Idleness and operational flexibility

To provide a more comprehensive understanding of how remote driving promote operational efficiency, we now analyze two additional performance measures: *idleness* and *operational flexibility*. Idleness is complementary to driver utilization and defined as the percentage of time drivers remain inactive during the planning horizon. Operational flexibility is quantified by the average number of *vehicle switches per driver*, i.e., the number of times a driver changes from one vehicle route to another. These metrics provide insights into the practical benefits and operational complexity of remote driver scheduling.

Fig. 7 presents a heatmap of average driver idleness across instance groups and workforce sizes. The results reveal substantial reductions in idleness levels for some instance groups, e.g., from 59.4% to 8.2% in group R1. Clustered instances (C1 and C2) consistently exhibit the highest idleness rates, exceeding 90% in most scenarios, which explains why these instances can be managed with significantly fewer drivers. For example, C2 instances maintain an idleness rate above 93% even with $H = 1$, indicating that a single driver can handle all driving tasks with minimal delay. In contrast, randomly distributed instances (R1 and R2) and mixed instances (RC1 and RC2) show lower idleness rates, particularly for restrictive workforce sizes. For R1 with $H = 2$, idleness drops to 8.2%, suggesting that drivers are almost continuously engaged in driving tasks, leaving little room for further workforce reductions. This observation is consistent with the infeasibility of R1 and RC1 instances for $H = 1$, as the limited idleness prevents assigning more tasks to a driver. Across all groups, average driver idleness generally decreases as H decreases, reflecting the consolidation of driving tasks among fewer drivers.

Fig. 8 illustrates the average number of driver switches as the workforce size decreases. As expected, the number of switches per driver increases substantially when fewer drivers are available. In this case, each driver must alternate among multiple vehicle routes to ensure all driving tasks are covered. As H decreases, the average number of switches per driver rises significantly, reaching averages of 20 or more in some groups (e.g., C1 and C2 with $H = 1$). Notably, the clustered instances (C1 and C2) exhibit the highest number of switches per driver for small H, which is consistent with their high idleness rates: the abundant idle time in these instances allows drivers to switch frequently between routes without violating time-window constraints. In contrast, the R1 and RC1 instances show lower switch counts, reflecting the limited idleness and tighter time windows that

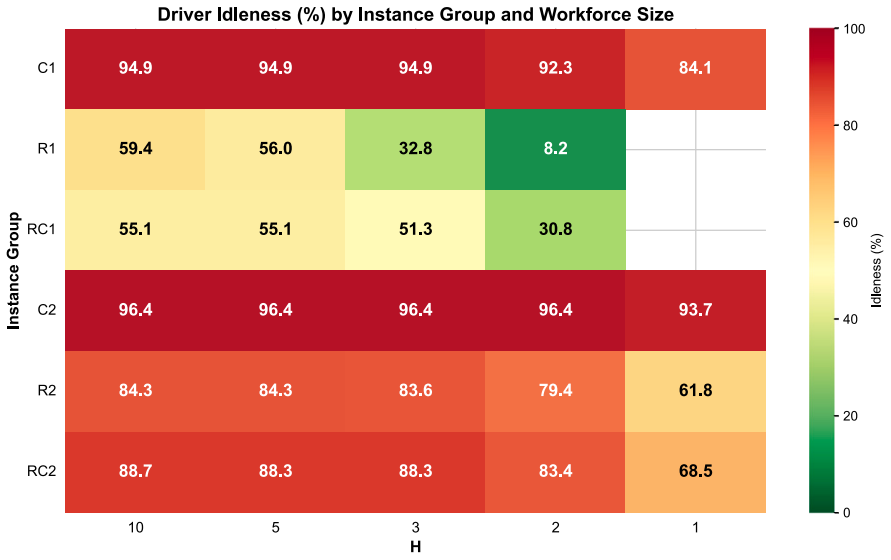


Fig. 7. Heatmap of average driver idleness (%) across instance groups and workforce sizes H. Higher values (darker red) indicate extended idleness and thus a higher potential for workforce reduction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

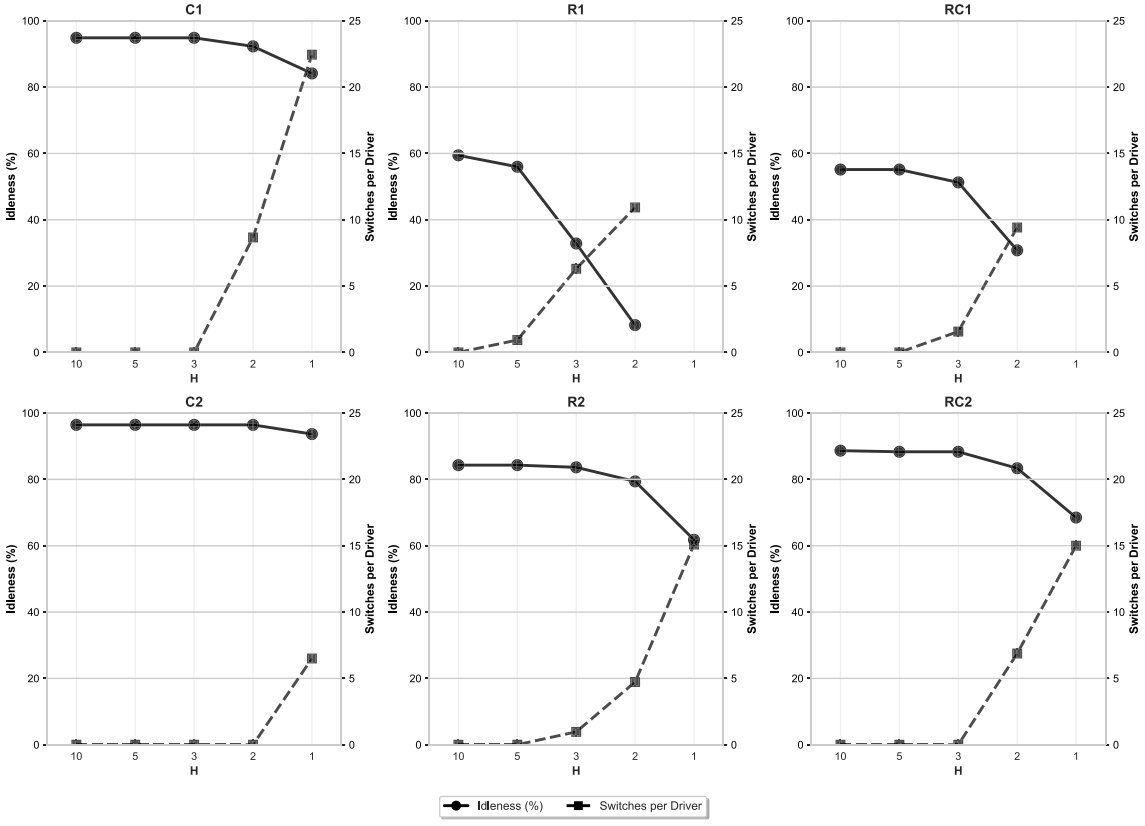


Fig. 8. Average driver idleness (left y-axis, solid line) and average number of switches per driver (right y-axis, dashed line) as a function of workforce size H across the six instance groups.

constrain driver flexibility. These findings highlight a fundamental tradeoff: while remote driving enables significant workforce reductions, it also increases operational complexity in terms of frequent driver switches.

Finding 2. The potential of remote driving to reduce the required driver workforce depends strongly on driver utilization. When driving legs are relatively short but time windows and service times are long, traditional on-board drivers may experience substantial idle time. Under such conditions, remote driving (particularly when setup times are short) can yield significant workforce reductions and thus economic benefits for logistics providers. However, the downsides of tighter remote-driver schedules should also be considered: reduced idle time and frequent switches between driving tasks may increase stress and

workload. Future research should therefore investigate the potential impacts of remote driving on driver well-being, system reliability, and road safety.

6. Conclusions

This paper introduces the Vehicle Routing Problem with Time Windows and Remote Drivers (VRPTWRD) which integrates decisions involving the allocation and scheduling of remote drivers and the design of least-cost vehicle routes. Remote driving has gained popularity. With an increasing number of in-house applications and on public roads, it is an important bridging technology until autonomous driving is the norm. Other than their on-board counterparts, remote drivers can switch among (even far-distant) vehicles without significant delay. Hence, the flexibility of remote driving promises a reduction in the driver workforce in times of driver shortages.

To model and solve the VRPTWRD for the first time, we propose a new compact formulation and branch-and-Benders-cut (BBC) methods based on Benders decomposition. Additionally, we present valid inequalities tailored for the VRPTWRD. The BBC methods differentiate the solvers for the Benders subproblem, either using a mixed-integer programming (MIP) or a constraint programming (CP) model. The results of computational experiments show that the BBC methods have a superior overall performance compared to solving the compact model with a general-purpose MIP solver, with a slight advantage for the BBC method that relies on the MIP model to solve the Benders subproblem. Moreover, the results indicate the relevance and benefits of integrating remote driver scheduling and vehicle routing. We observe a significant potential to reduce the driver workforce, compared to on-board driving and its one-to-one mapping between route and driver. In many instances, the optimal vehicle routes can be executed with two or three drivers fewer than the number of routes, indicating significant labor cost savings. In the vast majority of instances, reducing the workforce neither increases routing costs nor the number of routes. In the few instances with negative impact, there is just one additional route and at most a 10% increase in routing costs.

We believe our paper opens interesting opportunities for future research. First, alternative objective functions, other formulations, stronger valid inequalities, and more efficient solution approaches should be developed, especially relying on column generation algorithms. Second, the problem should be extended by additional technological (e.g., latency issues), driver well-being (e.g., fatigue effects after excessive remote driving without break), and legal (e.g., specific hours-of-service regulations for remote drivers) requirements as well as common routing extensions (e.g., multi-depot settings). Finally, case studies should validate the capability of remote driving based on real-world data.

CRediT authorship contribution statement

Pedro Munari: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Nils Boysen:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Pedro Munari thankfully acknowledges the funding of the São Paulo Research Foundation (FAPESP) [grant numbers 13/07375-0 and 22/05803-3]; the National Council for Scientific and Technological Development (CNPq) [grant numbers 405702/2021-3 and 314079/2023-8]; and Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) [Finance Code 001, PrInt 88887.837037/2023-00].

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ejor.2026.04.050.

References

Amador, O., Aramrattana, M., & Vinel, A. (2022). A survey on remote operation of road vehicles. *IEEE Access*, *10*, 130135–130154.

Benders, J. F. (1962). Partitioning procedures for solving mixed-variables programming problems. *Numerische Mathematik*, *4*(1), 238–252.

Boysen, N., & Fliedner, M. (2010). Cross dock scheduling: Classification, literature review and research agenda. *Omega*, *38*(6), 413–422.

Chao, I.-M. (2002). A tabu search method for the truck and trailer routing problem. *Computers & Operations Research*, *29*(1), 33–51.

Chen, J. Y., Haas, E. C., & Barnes, M. J. (2007). Human performance issues and user interface design for teleoperated robots. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, *37*(6), 1231–1245.

Codato, G., & Fischetti, M. (2004). Combinatorial Benders' cuts. In *Integer programming and combinatorial optimization: 10th international IPCO conference, New York, NY, USA, June 7-11, 2004. proceedings 10* (pp. 178–195). Springer.

Desaulniers, G., Madsen, O. B., & Ropke, S. (2014). The vehicle routing problem with time windows. In P. Toth, & D. Vigo (Eds.), *MOS/SIAM ser optim, Vehicle routing: problems, methods, and applications* (pp. 119–159).

Domínguez-Martín, B., Rodríguez-Martín, I., & Salazar-Gonzalez, J.-J. (2018). The driver and vehicle routing problem. *Computers & Operations Research*, *92*, 56–64.

Drexl, M. (2012). Synchronization in vehicle routing – A survey of VRPs with multiple synchronization constraints. *Transportation Science*, *46*(3), 297–316.

Garey, M. R., & Johnson, D. S. (1979). *Computers and intractability*: *vol. 174*, Freeman, San Francisco.

Garrido, F., & Resende, P. (2022). Review of decision-making and planning approaches in automated driving. *IEEE Access*, *10*, 100348–100366.

Gendron, B., Scutellà, M. G., Garroppo, R. G., Nencioni, G., & Tavanti, L. (2016). A branch-and-Benders-cut method for nonlinear power design in green wireless local area networks. *European Journal of Operational Research*, *255*(1), 151–162.

Goel, A. (2018). Legal aspects in road transport optimization in Europe. *Transportation Research Part E: Logistics and Transportation Review*, *114*, 144–162.

Gohar, A., & Lee, S. (2020). A cost efficient multi remote driver selection for remote operated vehicles. *Computer Networks*, *168*, Article 107029.

Guo, C., Sentouh, C., Haué, J.-B., & Popieul, J.-C. (2019). Driver-vehicle cooperation: A hierarchical cooperative control architecture for automated driving systems. *Cognition, Technology & Work*, *21*, 657–670.

Khan, M. A., Sayed, H. E., Malik, S., Zia, T., Khan, J., Alkaabi, N., & Ignatious, H. (2022). Level-5 autonomous driving – Are we there yet? A review of research literature. *ACM Computing Surveys*, *55*(2), 1–38.

Kim, J. (2006). A TCP/IP-based remote control system for yard cranes in a port container terminal. *Robotica*, *24*(5), 613–620.

Lam, E., Van Hentenryck, P., & Kilby, P. (2020). Joint vehicle and crew routing and scheduling. *Transportation Science*, *54*(2), 488–511.

Li, S., Zhang, Y., Blythe, P., Edwards, S., & Ji, Y. (2024). Remote driving as the failsafe: Qualitative investigation of users' perceptions and requirements towards the 5G-enabled level 4 automated vehicles. *Transportation Research Part F: Traffic Psychology and Behaviour*, *100*, 211–230.

Li, S., Zhang, Y., Edwards, S., & Blythe, P. T. (2023). Exploration into the needs and requirements of the remote driver when teleoperating the 5G-enabled level 4 automated vehicle in the real world – A case study of 5G connected and automated logistics. *Sensors*, *23*(2), 820.

Majstorović, D., Hoffmann, S., Pfab, F., Schimpe, A., Wolf, M.-M., & Diermeyer, F. (2022). Survey on teleoperation concepts for automated vehicles. In *2022 IEEE international conference on systems, man, and cybernetics* (pp. 1290–1296). IEEE.

Malik, S., Khan, M. J., Khan, M. A., & El-Sayed, H. (2023). Collaborative perception – The missing piece in realizing fully autonomous driving. *Sensors*, *23*(18), 7854.

Meisel, F., & Kopfer, H. (2014). Synchronized routing of active and passive means of transport. *OR Spectrum*, *36*, 297–322.

Moreno, A., Alem, D., Gendreau, M., & Munari, P. (2020). The heterogeneous multicrew scheduling and routing problem in road restoration. *Transportation Research Part B: Methodological*, *141*, 24–58.

Moreno, A., Munari, P., & Alem, D. (2019). A branch-and-Benders-cut algorithm for the crew scheduling and routing problem in road restoration. *European Journal of Operational Research*, *275*(1), 16–34.

Opiyo, S., Zhou, J., Mwangi, E., Kai, W., & Sunusi, I. (2021). A review on teleoperation of mobile ground robots: Architecture and situation awareness. *International Journal of Control, Automation and Systems*, *19*, 1384–1407.

Rahmaniani, R., Crainic, T. G., Gendreau, M., & Rei, W. (2017). The Benders decomposition algorithm: A literature review. *European Journal of Operational Research*, *259*(3), 801–817.

Rothenbächer, A.-K., Drexl, M., & Irnich, S. (2018). Branch-and-price-and-cut for the truck-and-trailer routing problem with time windows. *Transportation Science*, *52*(5), 1174–1190.

Soares, R., Marques, A., Amorim, P., & Parragh, S. N. (2024). Synchronisation in vehicle routing: Classification schema, modelling framework and literature review. *European Journal of Operational Research*, *313*(3), 817–840.

Society of Automotive Engineers (SAE) (2018). *Definitions for terms related to driving automation systems for on-road motor vehicles*: *vol. 3016*, Warrendale: SAE.

Solomon, M. M. (1987). Algorithms for the vehicle routing and scheduling problems with time window constraints. *Operations Research*, *35*(2), 254–265.

Teng, S., Hu, X., Deng, P., Li, B., Li, Y., Ai, Y., Yang, D., Li, L., Xuanyuan, Z., Zhu, F., et al. (2023). Motion planning for autonomous driving: The state of the art and future perspectives. *IEEE Transactions on Intelligent Vehicles*, *8*(6), 3692–3711.

Tilk, C., Bianchessi, N., Drexl, M., Irnich, S., & Meisel, F. (2018). Branch-and-price-and-cut for the active-passive vehicle-routing problem. *Transportation Science*, *52*(2), 300–319.

Wen, X., Sun, X., Sun, Y., & Yue, X. (2021). Airline crew scheduling: Models and algorithms. *Transportation Research Part E: Logistics and Transportation Review*, *149*, Article 102304.

Yu, Y., & Lee, S. (2022). Remote driving control with real-time video streaming over wireless networks: Design and evaluation. *IEEE Access*, *10*, 64920–64932.

Zhang, T. (2020). Toward automated vehicle teleoperation: Vision, opportunities, and challenges. *IEEE Internet of Things Journal*, *7*(12), 11347–11354.

Zulqarnain, S. Q., & Lee, S. (2021). Selecting remote driving locations for latency sensitive reliable tele-operation. *Applied Sciences*, *11*(21), 9799.